

SAI SRIKAR REDDY KOLLI

+1(573)801-6044 • srikarreddy0805@gmail.com • LinkedIn • GitHub

EDUCATION

Computer and Information Sciences

University of Missouri, Columbia

Aug 25 - Present

4.0 GPA

Department of Computer Science Engineering

Indian Institute of Information Technology, SriCity

Sep 20 - Aug 24

7.64 GPA

TECHNICAL SKILLS

Programming Databases: Python, R, Java, C, C++, JavaScript (TypeScript, Node.js), SQL, PostgreSQL, MongoDB, GraphQL

Web Application Development: ReactJS, Angular, HTML, CSS, Flask, Django, Spring Boot, PyTest, Selenium

Cloud DevOps: AWS (EC2, S3, IAM, ACM, Lambda, API Gateway, Step Functions, Amplify), Azure DevOps, Docker, Kubernetes, Ansible, Jenkins, GitLab CI, Prometheus, Grafana

Tools and Platforms: Git, Power BI, Databricks, Azure, MATLAB, Prophecy, Microsoft Excel, Power Apps

EXPERIENCE

University of Missouri Columbia: Graduate Student Assistant

Aug 25 - Present

- Operationalized **IBM Qiskit** Variational Quantum Circuits within a **PyTorch** workflow, utilizing Hilbert space projection to solve complex class separability issues in NISQ-era environments
- Architected a Hybrid **ResNet-GNN-Quantum (HRGQ)** framework that achieved 98.4% accuracy on **CIC-IoT-2023**, outperforming classical baselines in reducing false positives.
- Developing an intelligent document processing pipeline that allows users to upload and query files (PDF, TXT, CSV) using **RAG (Retrieval-Augmented Generation)** techniques.

MAQ Software: Software Engineer 1

May 24 - Jun 25

- Analyzed legacy data workflows and refactored **SQL/PySpark** pipelines in **Azure ADF/Databricks**, improving data quality by 10% and enabling cloud-ready architecture for scalable processing.
- Developed end to end workflow diagrams, semantic models, and automated documentation for **ETL** and **Power BI**, reducing onboarding time by 30% and improving reporting consistency across teams.
- Architected scalable ETL pipelines in **Azure Data Factory (ADF)**, establishing Linked Services to ingest data from 7 distinct heterogeneous sources into **Azure Data Lake Storage (ADLS)**, serving as the central **OneLake** repository for downstream analytics

SP Software (P) Limited: Associate Software Engineer Intern

Feb 24 - Apr 24

- Enhanced image processing pipeline by integrating **ESRGAN** algorithms, achieving a 5% improvement in text clarity and legibility, while decreasing image processing time by 15 milliseconds per image
- Conducted research to explore cutting-edge methods for enhancing text visibility and quality in images using **ESRGAN** techniques.

ACADEMIC PROJECTS

Ticketmaster

Github [↗](#)

- Engineered a robust seat reservation engine handling 10,000+ concurrent requests by implementing a hybrid locking strategy. Implemented atomic batch processing for the real-time seat map, eliminating booking conflicts
- Reduced database read load by 60% and API latency by 200ms by implementing a Write-Through Caching strategy with Redis for real-time seat availability maps.

GraphNet-Sentry

Github [↗](#)

- Developed a unified deep learning framework capable of processing heterogeneous data modalities, successfully applying ResNet-50 + Graph Neural Networks (GNN) to both Network Intrusion Detection and Satellite Image Analysis.
- Engineered a "Tabular-to-Image" transformation engine, enabling the model to achieve 99.9% accuracy on the CICIoT-2023 security dataset by treating network flows as visual patterns.

CERTIFICATIONS

Microsoft: Microsoft Certified: Fabric Analytics Engineer Associate

Certification [↗](#)

DeepLearning.AI: Supervised Machine Learning: Regression and Classification

Certification [↗](#)

DeepLearning.AI: Neural Networks and Deep Learning

Certification [↗](#)